Nvidia vs AMD

Executive Summary

This report evaluates NVIDIA and Advanced Micro Devices (AMD) to determine which company offers the stronger long-term investment opportunity. The analysis combines qualitative company research with quantitative financial modelling, including discounted cash flow valuation, historical stock performance, technical analysis using moving averages, and a moving average crossover trading strategy.

Both companies operate within the rapidly expanding semiconductor industry and are positioned to benefit from increasing demand for artificial intelligence , cloud computing and data centre infrastructure. However, NVIDIA currently demonstrates stronger financial performance, higher profitability and greater market leadership.

The DCF valuation, historical market performance and technical analysis all suggest that NVIDIA represents the stronger long-term investment despite its higher valuation.

Semiconductor Industry Overview

The semiconductor industry has become one of the fastest growing sectors in the global economy. Demand for advanced processors has spiked due to major developments in artificial intelligence, machine learning, autonomous vehicles and cloud computing.

Companies operating within the industry benefit from an array of long-term growth drivers such as; AI, demands for new data centres, expansion of cloud computing, Commercial use in gaming and High-performance computing.

However, despite these opportunities, the industry's stability remains at risk due to potential supply chain disruptions.

Company overview

NVIDIA

NVIDIA is the global leader in graphics processing units and AI accelerators. While originally focused on gaming GPU's, the company now focuses on AI infrastructure and date centres, which generates majority of their revenue.

Advantages

- Market leader in AI chips
- Strong pricing power
- Exceptional operating margins

AMD

AMD has transformed into NVIDIA’s largest competitor through significant improvements in processor and graphics technology.

AMD competes across

- CPU’s
- GPU’s
- Data centre processors

Although AMD has also experienced major revenue growth, it remains smaller than NVIDIA and faces greater competitive pressure.

Financial Analysis

Both companies have demonstrated strong financial performances over recent years, however NVIDIA has consistently generated more profit compared to AMD

- NVIDIA outperforms AMD in revenue growth, with almost 115% growth whereas AMD’s 34% growth in the 2025 fiscal year.
- NVIDIA has a higher Gross margin with a 71% gross margin, compared to AMD’s 51%.
- NVIDIA outperforms in operating margins with a staggering 60% when compared to 10% from AMD

In addition, NVIDIA has a greater pricing power as they are one of few companies in their industry capable of supplying the enormous demands generated by the AI boom.

DCF Valuation

NVIDIA

year	FCF $bn	DCF $bn	assumed Growth	TV $bn	Dr %
2026	131.283	119.359032	60%	249.939	9.99
2027	170.6679	141.073499	30%		
2028	204.80148	153.912355	20%		
2029	231.4256724	158.124340	13.%		
2030	249.9397262	155.263466	8%		
Total DCF $bn		882.996145			

WACC:Nvidia	Market cap $bn	Total Debt $bn	Total Capital $bn	cost of equity %	cost of debt %	Corporate tax rate %
9.99%	5000	9.9	206	10	4	13

AMD

year	FCF $bn	DCF $bn	assumed Growth	TV $bn	Dr %
2026	48.44	44.0483768	40%	86.4247	9.97
2027	60.55	50.0686285	25%		
2028	71.449	53.7246357	18%		
2029	80.02288	54.7163699	12%		
2030	86.4247104	53.7361821	8%		
Total DCF $bn		310.030375			

WACC:AMD	Market cap $bn	Total Debt $bn	Total assets $bn	cost of equity %	cost of debt %	Corporate tax rate %
9.97%	790	3.87	79	10	4.3	14

A discounted cash flow model was constructed for both companies using forecast revenue growth, operating margins and terminal value assumptions.

The model estimates the intrinsic value of each company by discounting future cash flows back to present value.

For Revenue growth I assumed that for both companies would experience an increase revenue however over time growth would decrease as the demand in the semiconductor market stabilises.

Based on these assumptions, NVIDIA produced the higher intrinsic valuation, supported by stronger expected earnings growth and superior profitability.

Although valuation models remain highly sensitive to assumptions, the analysis suggests NVIDIA possesses greater long-term value creation potential.

Python Analysis

Using the yfinance library in python I was able to conduct analysis of both AMD and NVIDIA stocks.

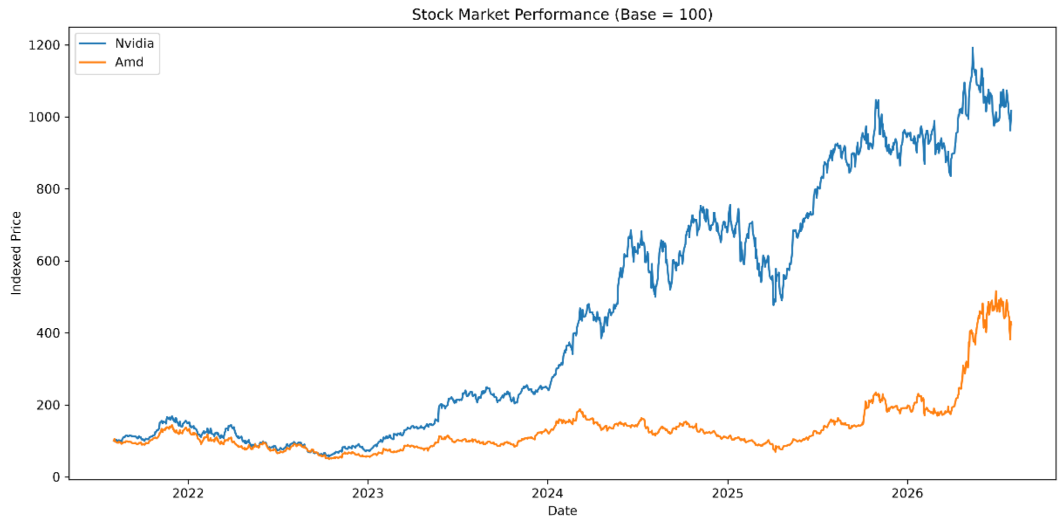
This includes

- Historical market performance
- 50- day moving average
- 200-day moving average
- Annual return
- Annual volatility
- Correlation
- Sharp ratio
- Moving average trading strategy

Market performance

Normalised stock prices demonstrate that NVIDIA has significantly outperformed AMD over the analysis period.

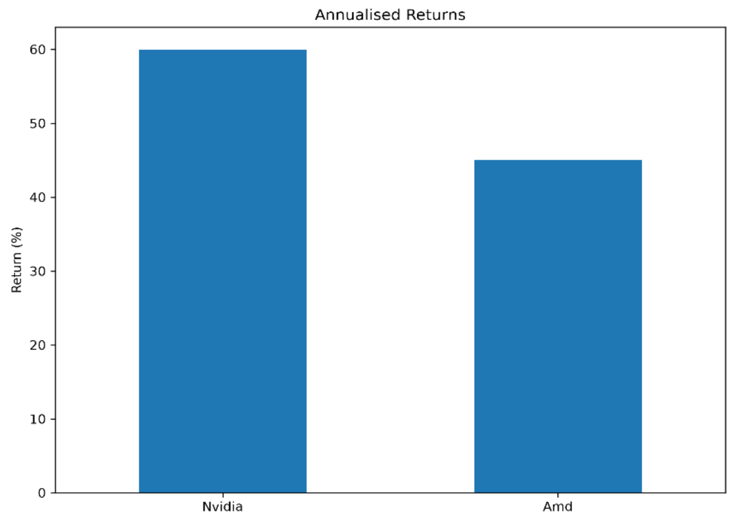
This reflects investors' confidence in NVIDIA's dominant position within the AI semiconductor market.



Annual Return

Historical return analysis indicates that NVIDIA generated substantially higher annual returns than AMD, With NVIDIA producing 60% annual returns compared to only 45% by AMD

This reflects stronger earnings growth and greater investor demand for NVIDIA shares during the AI investment boom.

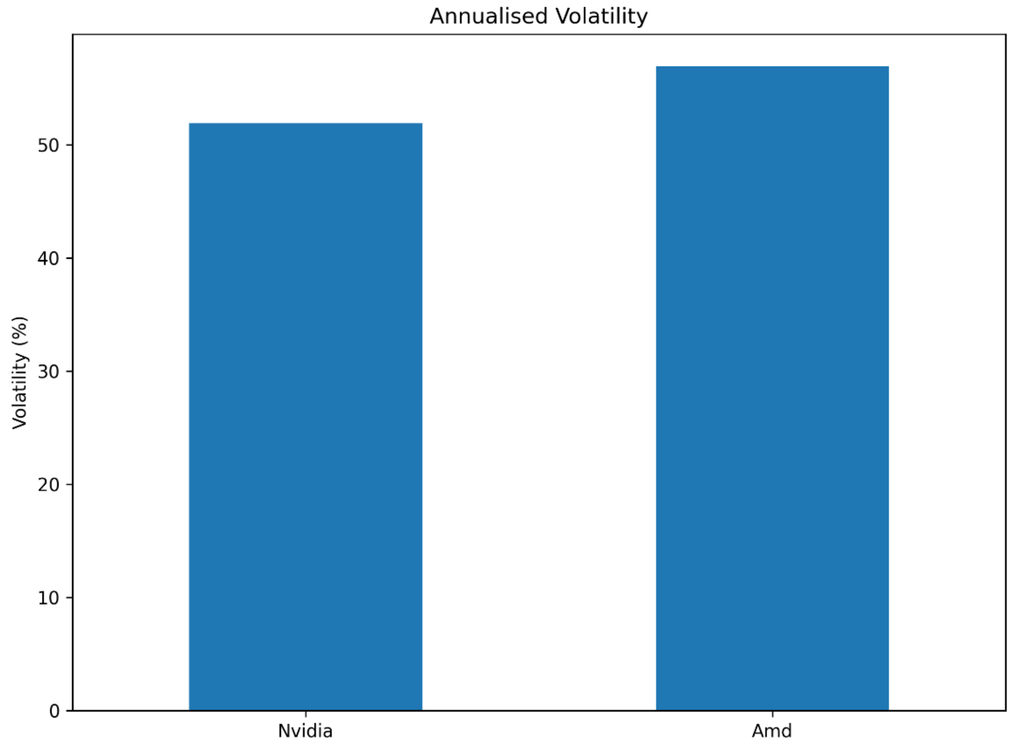


Annual Volatility

Both companies exhibit relatively high volatility, reflecting the cyclical nature of semiconductor businesses.

AMD demonstrated slightly higher volatility, 57% compared to NVIDIA's 52% , indicating larger price fluctuations and therefore higher investment risk.

Investors should consider this increased risk when evaluating expected returns.



Correlation

he correlation matrix was calculated to examine the relationship between NVIDIA and AMD's daily stock returns. The analysis produced a correlation coefficient of **0.662**, indicating a moderately strong positive relationship between the two companies.

A positive correlation suggests that both stocks generally move in the same direction, which is expected as NVIDIA and AMD operate within the semiconductor industry and are influenced by similar macroeconomic and industry-specific factors, including demand for artificial intelligence, gaming hardware and data centre infrastructure.

However, the correlation is well below **1.0**, meaning the stocks do not move identically. Company-specific factors such as product launches, earnings announcements and changes in market share can cause their performances to diverge. Consequently, investors may still gain some diversification benefits by holding both companies within the same portfolio.

Sharp Ratio Analysis

The Sharpe ratio was calculated to evaluate the risk-adjusted performance of each investment. Unlike annual returns alone, the Sharpe ratio measures how much return an investor receives for each unit of risk taken.

The results were:

Company	Sharpe Ratio
NVIDIA	1.07
AMD	0.71

A higher Sharpe ratio indicates a more attractive balance between return and volatility. NVIDIA achieved the higher Sharpe ratio of 1.07, suggesting that it generated stronger returns relative to the level of risk undertaken by investors. In comparison, AMD's Sharpe ratio of 0.71 indicates that, although it produced positive returns, investors received less return for each unit of risk.

These findings reinforce the conclusions from the annual return and volatility analysis. While both companies have benefited from the rapid expansion of the AI semiconductor industry, NVIDIA has delivered superior risk-adjusted performance over the analysis period, making it the more attractive investment based on historical market data.

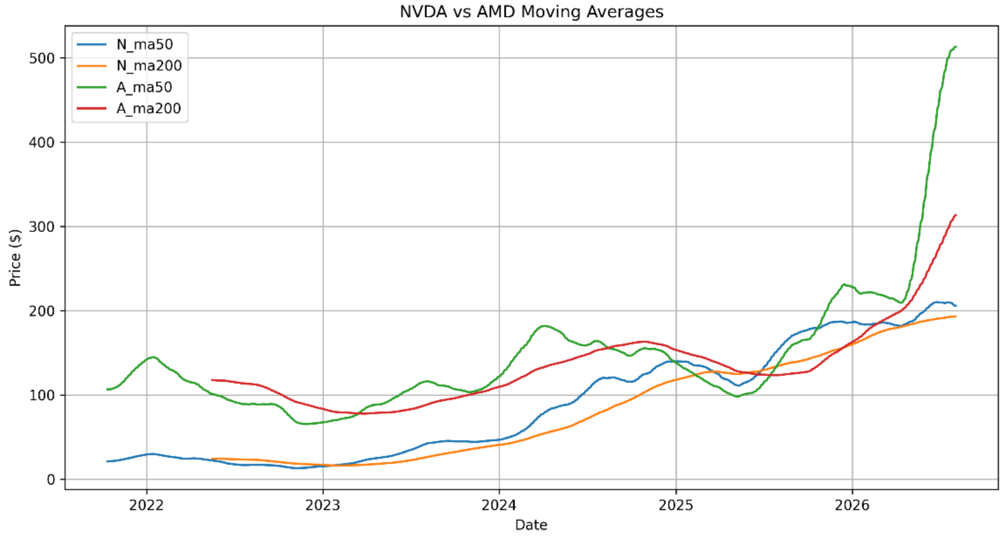
Technical Analysis

A 50-day and 200-day moving average were calculated to identify long-term price trends.

Periods where the 50-day moving average crossed above the 200-day moving average (Golden Cross) generated buy signals.

When the 50-day moving average crossed below the 200-day moving average (Death Cross), sell signals were generated.

Overall, NVIDIA displayed stronger long-term upward momentum than AMD.



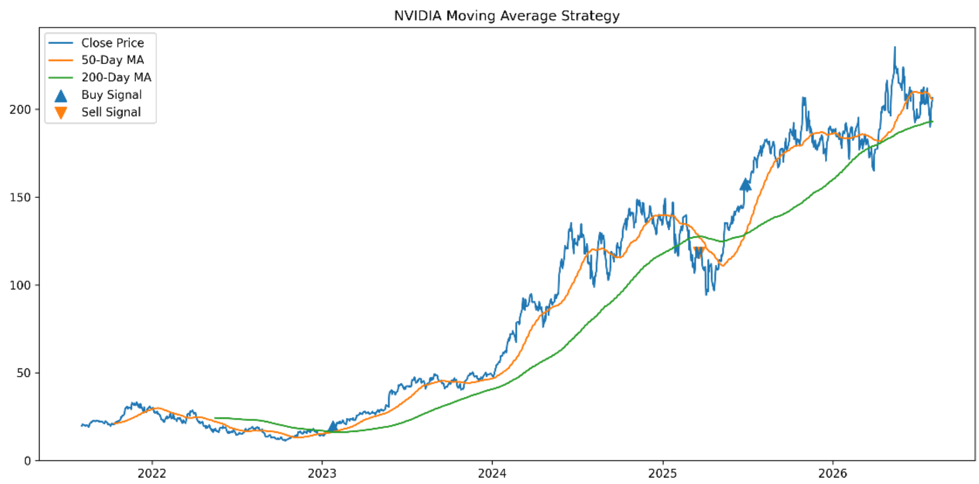
Trading Strategy

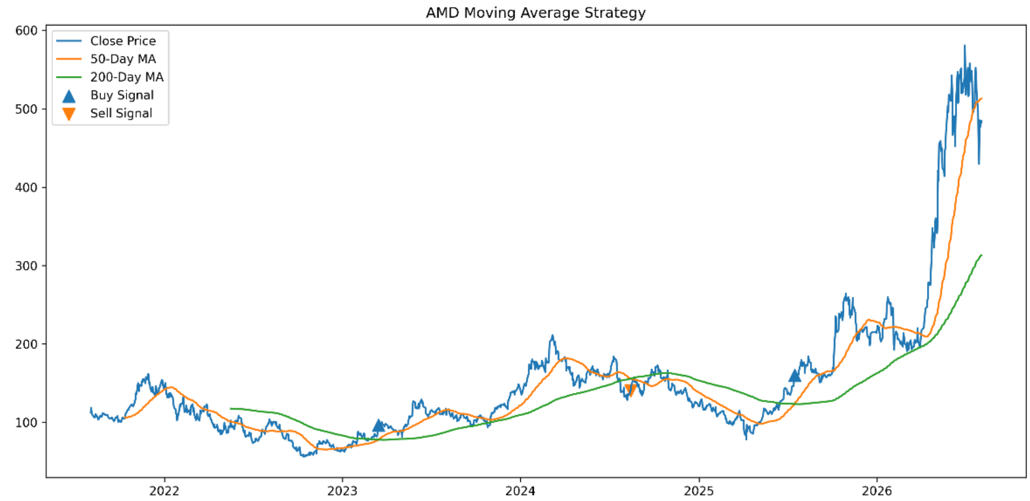
A moving average crossover strategy was implemented in Python.

Trading Rules

- Buy: 50-Day Moving Average crosses above the 200-Day Moving Average.
- Sell: 50-Day Moving Average crosses below the 200-Day Moving Average.

- Each strategy began with an initial investment of **$1,000**.
- The Python model automatically simulated trades based on these signals and calculated the final portfolio value.
- The strategy successfully captured major long-term trends while avoiding periods of prolonged market weakness.
- Although simple, this demonstrates how technical indicators can be incorporated into systematic investment strategies.





Risk

Despite NVIDIA's attractive outlook, several risks remain.

- **Competition:** AMD continues investing heavily in AI processors and could reduce NVIDIA's market share
- **Valuation:** NVIDIA currently trades at a relatively high valuation compared with historical averages. If future earnings disappoint investors, the share price could decline significantly.
- **Export Restrictions:** Government restrictions on semiconductor exports could reduce sales in key international markets.
- **Technological Changes:** Rapid innovation within the semiconductor industry requires continuous research and development spending.

A major risk is the chance of supply chain shortages. A key example is the silicon shortage experienced between 2021 and 2023. Following the initial lockdown due to covid-19 the demand for consumer electronics increased exponentially, however with manufacturing plants operating at reduced speed and the lack of raw silicon, the extreme demand caused the price of silicon to spike by almost 300%. Over 169 industries were affected and semiconductor experienced little to no revenue growth between 2022 and 2023 fiscal years, with NVIDIA growing by only 0.33% and AMD losing revenue with a 3.8% decrease. Since this era in semiconductor manufacturing, companies like TSMC and Samsung have invested in manufacturing facilities however these take years to become fully operational and with the current demand for new products, another supply chain problem would be detrimental to the stability of the semiconductor market

Final summary

Metric	**NVIDIA**	**AMD**	**Better Performer**
Annualised Return	**60.5%**	45.4%	NVIDIA
Annualised Volatility	Lower	Higher	NVIDIA
Sharpe Ratio	**1.07**	0.71	NVIDIA
DCF Valuation	Higher intrinsic value	Lower intrinsic value	NVIDIA
Trading Strategy Return	$7846.56	$4333.89	*NVIDIA*
Overall Recommendation	**BUY**	HOLD	NVIDIA

Final Recommendations

After combining qualitative research with quantitative analysis, NVIDIA appears to offer the stronger long-term investment opportunity.

The company benefits from:

- Market leadership in AI hardware
- Strong revenue growth
- Superior profitability
- Higher intrinsic value from the DCF model
- Strong historical market performance
- Positive long-term technical indicators

Although AMD remains an attractive growth company with significant future potential, NVIDIA currently demonstrates stronger competitive advantages and financial performance.

Final Recommendation: BUY NVIDIA

Appendix

Python Skills Demonstrated

- Financial data collection using yfinance
- Data manipulation using pandas
- Data visualisation using matplotlib
- Return calculations
- Volatility calculations
- Moving average analysis
- Trading strategy back testing

Financial Skills Demonstrated

- Equity Research
- Industry Analysis
- Financial Statement Analysis
- Revenue Forecasting
- Discounted Cash Flow Valuation
- Technical Analysis
- Investment Recommendation